



Using Pull Requests to Make Collaboration Visible in CDIO Project-Based Courses

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INDUSTRIAL ENGINEERING

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Motivation



Project-based courses

Assessment often prioritizes final artifacts, limiting visibility into collaboration, reasoning, and responsiveness to feedback

Professional practice

Modern technical work relies on version control, peer review, and documented iteration—skills students are expected to use immediately in the workforce

Assessment misalignment

What we value in learning is not always what we can currently observe

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└ Motivation

Project-based courses are widely used – and we care about **collaboration**, **reasoning**, and **responsiveness to feedback**.

But in practice, assessment still prioritizes **final artifacts** – reports, code, presentations.

At the same time, professional work is inherently **collaborative** and **iterative**.

Version control, peer review, and documented iteration are not optional – they are core practice.

So the tension is this – what we value in learning is not always what we can actually **observe and assess**.

Motivation
 Project-based courses Assessment often prioritizes final artifacts, limiting visibility into collaboration, reasoning, and responsiveness to feedback
 Professional practice Modern technical work relies on version control, peer review, and documented iteration—skills students are expected to use immediately in the workforce
 Assessment misalignment What we value in learning is not always what we can currently observe

Framing the problem



Core question

How can collaboration, accountability, and responsiveness to feedback be made **visible and assessable** in team-based projects?

- ▶ final deliverables hide process and contribution
- ▶ peer interaction is difficult to audit retrospectively
- ▶ feedback often resets instead of accumulating

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└ Framing the problem

This leads to the core question –

How do we make **collaboration**, **accountability**, and **responsiveness to feedback** visible and assessable?

In team projects, the final deliverable hides most of what matters – who did what, how decisions were made, and how feedback was handled.

Peer interaction is difficult to reconstruct after the fact – and feedback often **resets** instead of accumulating.

So the issue is not lack of activity – it is lack of **visibility**.

Not just another tool

Git/GitHub is an industry-standard collaboration infrastructure

- ▶ used across software, data science, and engineering teams
- ▶ supports version control, review, and reproducible handover
- ▶ directly transferable – students can credibly list this on their CV

🌟 **Motivation: students practice skills they will immediately use**

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└ Learning with professional collaboration tools

The approach is to use professional collaboration tools – not as an add-on, but as part of the learning environment.

Git and GitHub are widely used across technical fields.

They support **version control**, **peer review**, and **reproducibility**.

Importantly, these are skills students are expected to use immediately after graduation.

So this is not just about authenticity – it is about **transfer**. Students practice what they will actually use.

Minimal GitHub concepts (for assessment)



Repository

A shared workspace containing code, data, documentation, and history

Pull Request (PR)

A structured proposal to change shared work, including:

- ▶ rationale and scope
- ▶ peer review and discussion
- ▶ explicit approval and merge decision

 **Key idea: PRs bind technical change to collaborative process**

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└ Minimal GitHub concepts (for assessment)

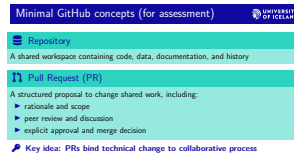
Briefly aligning terminology –

A repository is a shared workspace containing code, data, and history.

A pull request – or PR – is a structured proposal to change shared work.

It includes a rationale, and is reviewed, discussed, and approved before merging.

The key idea is that PRs bind **technical change** to **collaborative process**.



Design rationale



Starting point

- ▶ CDIO-aligned
- ▶ assessment as development
- ▶ observable collaborative

Design claim

Pull requests are not treated as a tooling detail, but as **assessment infrastructure**:

 feedback +  iteration $\Rightarrow$  competence

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└ Design rationale

The design is CDIO-aligned and treats assessment as a **developmental process**.
The focus is on **observable collaboration**, not just outcomes.
The central claim is that pull requests are not just a tool – but can function as **assessment infrastructure**.
They make feedback and iteration visible – and over time this supports competence development.

Design rationale



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Why pull requests matter



Repository

A shared workspace containing code, data, documentation, and history

Pull Request (PR)

A structured proposal to change shared work through:

- ▶ rationale and scope
- ▶ peer review and discussion
- ▶ approval and merge decision

Why it matters for assessment

- ▶ reasoning becomes visible
- ▶ contribution becomes attributable
- ▶ critique and negotiation are captured
- ▶ responses to feedback are auditable

Reframing

PRs expose the learning process

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└ Why pull requests matter

A PR naturally contains what we want to assess – rationale, critique, discussion, and decisions.

This means reasoning becomes **visible**, contributions become **attributable**, and responses to feedback become **auditable**.

So the reframing is simple – PRs expose the learning process.

Why pull requests matter	
 Repository A shared workspace containing code, data, documentation, and history	 Why it matters for assessment ▶ reasoning becomes visible ▶ contribution becomes attributable ▶ critique and negotiation are captured ▶ responses to feedback are auditable
 Pull Request (PR) A structured proposal to change shared work through: ▶ rationale and scope ▶ peer review and discussion ▶ approval and merge decision	 Reframing PRs expose the learning process

Assessable behaviors made visible

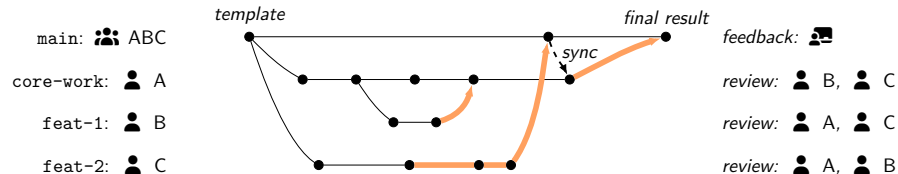


Required practices

- ▶ focused PRs with clear intent
- ▶ at least two substantive reviews
- ▶ explicit responses to all comments
- ▶ merge only after review

Observable evidence

- ▶ authorship and co-authorship
- ▶ depth of critique and follow-up
 - ▶ actionable comments (no LGTM or 🍌)
- ▶ handover quality
- ▶ documentation updates



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Assessable behaviors made visible

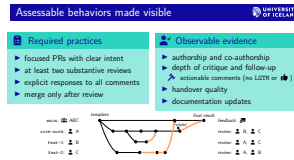
We define specific required practices.

Students submit focused PRs, receive at least two **substantive** reviews, and respond explicitly to all comments.

Substantive means actionable feedback – not just superficial approval

Assessment then uses observable evidence – authorship, critique, follow-up, and handover.

So assessment shifts from hidden process to **visible artifacts**.



Scaffolded progression across courses



Information Engineering (IE)

- ▶ 20–25 students
- ▶ 6–7 small teams
- ▶ early undergraduate
- ▶ new repository each cycle

Main purpose

Tooling and collaboration practices are the learning focus

Business Intelligence (BI)

- ▶ 10–15 students
- ▶ 2 large teams
- ▶ assumes GitHub fluency
- ▶ single persistent repository

Main purpose

Tooling supports more advanced project work

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└ Scaffolded progression across courses

This is implemented across two courses.

In Information Engineering, the focus is learning the tools – small teams, repeated practice, new repositories.

In Business Intelligence, students are assumed fluent – one persistent repository across the semester.

So there is progression – from learning the workflow to using it as **infrastructure**.

Scaffolded progression across courses	
Information Engineering (IE)	Business Intelligence (BI)
▶ 20–25 students ▶ 6–7 small teams ▶ early undergraduate ▶ new repository each cycle	▶ 10–15 students ▶ 2 large teams ▶ assumes GitHub fluency ▶ single persistent repository
Main purpose Tooling and collaboration practices are the learning focus	Main purpose Tooling supports more advanced project work

Feedback carry-forward



Cumulative assessment

- ▶ unresolved PR comments remain active
- ▶ later work is checked against prior feedback
- ▶ capstone evaluation includes earlier issues

Effect

- ▶ continuity
- ▶ accountability
- ▶ coherence
- ▶ less fragmentation

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└ Feedback carry-forward

When teams are working with a single, persistent repository, feedback can carry forward – it doesn't reset between assignments.

If an issue is raised in a pull request and not resolved, it remains visible in the repository. Later work is then evaluated in light of that earlier feedback – including in the capstone.

So feedback accumulates over time rather than fragmenting across tasks.

This creates **continuity** and **accountability**.

Students cannot simply move on – they have to revisit and resolve issues.

And that is much closer to professional practice.

 Cumulative assessment	 Effect
▶ unresolved PR comments remain active ▶ later work is checked against prior feedback ▶ capstone evaluation includes earlier issues	▶ continuity ▶ accountability ▶ coherence ▶ less fragmentation

Indicative evidence



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Indicative evidence

Indicative evidence		
Outstanding	Good	Developmental
GitHub usage Branches are well organized with logical names tied to features. Commits are coherent and clearly describe changes. Documentation is complete and provides clear instructions. Teamwork All team members actively contribute. Contributions and responsibilities are distributed across the team. Decisions are shared through review and discussion.	Branches are used but naming or scope is sometimes inconsistent. Commits are mostly coherent with occasional issues. Documentation is present but incomplete. Most members participate, but contributions are uneven. One or two members lead development, while others contribute through reviews or comments.	Branches are absent or unclear. Commits are inconsistent with little repetition. Documentation is unclear or missing. Work is largely carried out by one individual. Participation is restricted or uneven across the team.
Calibration ► levels: developmental → outstanding ► same dimensions, increasing expectations ► signals expectations for Capstone		Observed trend ► GitHub usage improved over repeated cycles ► teamwork & participation improved more strongly

Calibration

- levels: developmental → outstanding
- same dimensions, increasing expectations
- signals expectations for Capstone

Observed trend

- GitHub usage improved over repeated cycles
- teamwork & participation improved more strongly

To clarify the rubric design –

The GitHub usage and teamwork dimensions are **shared across both courses**. Early on, we are deliberately **lenient** – students are still learning the workflow.

We observe improvement in GitHub use and stronger improvement in teamwork and participation. Qualitatively, PRs become more focused, better documented, and more responsive. In the BI course, each thematic module also has its own rubric – and these are **reused directly in the capstone**.

During the learning phase, grading is **normalized to “very good”**. So students are not penalized for early-stage work –

but they are explicitly shown what will be expected in a more **refined, final version**.

This allows them to **prioritize effort** – they can revisit earlier work, address feedback, and align with final expectations in the capstone.

So students are not only evaluated – they are shown what “good enough” looks like **before** the capstone.

Balancing team and individual assessment



Team-level

- ▶ analytical quality
- ▶ coherence
- ▶ shared responsibility

Individual-level

- ▶ visible contributions
- ▶ review quality
- ▶ responsiveness to feedback

Design goal

Accountability without fragmenting teamwork

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└ Balancing team and individual assessment

A common concern is balancing team and individual assessment.

Team-level outcomes still matter – quality and coherence.

But individual contribution becomes visible through artifacts – commits, reviews, and discussion.

The goal is **accountability** without breaking **collaboration**.

Team-level	Individual-level
▶ analytical quality ▶ coherence ▶ shared responsibility	▶ visible contributions ▶ review quality ▶ responsiveness to feedback

 Design goal
Accountability without fragmenting teamwork

Key shifts in practice



 **Quality**
iteration
vs. reviewable work

 **Collaboration**
implicit teamwork
vs. explicit review

 **Workflow**
informal work
vs. structured process

Takeaway

Making collaboration visible turns implicit practices into something we can teach and assess

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└─ Key shifts in practice

What PR-based workflows do is make collaboration practices **explicit**.

Students are not used to reviewing each other's work critically. That's not a problem – it's something we need to **teach and normalize**.

Similarly, work becomes visible much earlier. Students can no longer rely on polishing something at the end – they have to produce work that can be reviewed along the way.

And the workflow itself introduces structure. That can feel like overhead, but it is also what enables learning – especially around collaboration and feedback.

So rather than thinking of these as problems or tensions, they are **practices becoming visible** – and therefore teachable.

Key shifts in practice



Quality	Collaboration	Workflow
iteration vs. reviewable work	implicit teamwork vs. explicit review	informal work vs. structured process

 **Takeaway**
Making collaboration visible turns implicit practices into something we can teach and assess

Limitations and next steps



Boundary conditions

- ▶ design-oriented, not experimental
- ▶ depends on student readiness
- ▶ larger cohorts require more coordination

Next iterations

- ▶ extend to more courses
- ▶ structured team-building to accelerate trust
- ▶ refined review calibration

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└─ Limitations and next steps

There is definitely some initial overhead – especially around learning the workflow and review practices.

Students are not used to this level of structure or visibility.

It also depends on readiness – this works much better once basic tooling is already in place.

And scaling requires coordination, particularly around review quality.

So this is not a drop-in solution – it needs to be designed into the course.

Limitations and next steps



Boundary conditions

- ▶ design-oriented, not experimental
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Next iterations

- ▶ extend to more courses
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- ▶ refined review calibration

Conclusions



Key takeaway

PR-based assessment makes collaboration, accountability, and responsiveness to feedback visible and assessable

- ▶ assessment moves from final products to visible process
- ▶ feedback carries forward instead of resetting
- ▶ students practice professional collaboration in routine course work

Contribution

This is an assessment architecture, not just a tool choice

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Conclusions

The key takeaway is that making collaboration visible fundamentally changes what we can assess.

There is an initial overhead – students don't always enjoy the structure at first. Reviewing peers, documenting work, responding to feedback – this is unfamiliar and effortful.

But over time, the quality of work improves significantly. Work becomes more structured, better justified, and more coherent.

And importantly, students recognize this afterwards. Many of them explicitly say that this is where they learned the most – not just the content, but how to actually work.

So the impact is not just in what they produce – it's in how they approach collaboration, feedback, and responsibility.

And that comes from the **assessment structure**, not the specific subject matter.

Conclusions

Key takeaway

PR-based assessment makes collaboration, accountability, and responsiveness to feedback visible and assessable

- ▶ assessment moves from final products to visible process
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Contribution

This is an assessment architecture, not just a tool choice

Questions?



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Questions?

A few questions that often come up –

Isn't this a lot of overhead? Yes – there is initial overhead. But it decreases quickly, and the payoff is that assessment becomes much easier and more transparent later.

What about students with no GitHub experience? We scaffold this earlier in the curriculum. In later courses, we assume fluency so the tooling does not dominate the learning.

Does this scale? It can scale, but requires coordination – especially around review practices. The visibility actually helps manage larger groups.

How do you ensure fairness in team assessment? Individual contributions are visible through PRs, reviews, and discussion. So we don't rely only on peer evaluation – we have **artifact-based evidence**.

Is this specific to software or data courses? The tooling is domain-specific, but the principle is not. The idea is to make process and feedback visible – the same logic can apply elsewhere.

Is this just about GitHub? No – GitHub is the **enabler**. The contribution is the **assessment architecture**.

